

Fractional Reservoir Computing for Critical Brain Dynamics: A Review of the Mathematical and Modelling Foundations

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Companion document to the **fracres** library

September 2026

Abstract

The **fracres** library builds reservoir computers whose nodes carry fractional (power-law, non-Markovian) memory, driven by fractional Gaussian noise and regularised toward critical dynamics in a Besov/ L^p geometry, with the aim of generating and analysing synthetic macroscopic neural signals with realistic long-range dependence (LRD) and critical avalanche statistics. This review surveys the four bodies of theory the construction rests on — (i) fractional calculus and the stability of fractional-order systems, (ii) reservoir computing and the echo-state/edge-of-chaos literature, (iii) neural-mass and neural-field models and their fractional generalisations, and (iv) critical brain dynamics, neuronal avalanches, self-organised and quasi-self-organised criticality with homeostatic control — together with the mathematics of long-range dependence: self-similar processes, heavy tails, and wavelet/Besov estimation. For each area we state the definitions and theorems the library uses, summarise the neurophysiological evidence, and identify the open gaps the **fracres** programme is positioned to probe. All references were verified programmatically against publisher/Crossref, OpenAlex, or arXiv records in September 2026.

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1 Introduction and scope

Macroscopic neural signals — EEG, MEG, local field potentials — exhibit two statistical signatures that any generative model of brain dynamics must confront: *long-range dependence* (power-law temporal correlations and $1/f$ -type spectra persisting over hundreds of seconds [31, 41]) and *criticality* (power-law distributions of neuronal avalanche sizes and durations, with exponents near the mean-field branching values $\tau \approx 1.5$ and $\alpha \approx 2.0$ [6, 14]). Classical generative models struggle with both: reservoir computers forget exponentially [35], neural-mass models relax exponentially [36], and strict self-organised criticality requires conservation laws that brains do not obey [11].

The **fracres** project takes the position that these failures share a common mathematical root: the dynamics are built from *integer-order* (exponential-memory) operators, while the physics of neural tissue — from single-neuron adaptation [45] to population filtering — is better described by *fractional-order* (power-law memory) operators. The library therefore places long memory in the *nodes* of a reservoir computer via discrete fractional derivatives, drives the system with fractional Gaussian noise of tunable Hurst exponent, stabilises it at the fractional edge of chaos via Matignon’s stability criterion, regulates it with a quasi-self-organised-criticality (qSOC) homeostat, and constrains its outputs to the Besov-regularity manifold that heavy-tailed long-memory signals inhabit. We call the resulting generative models *phantom brains*: synthetic neural signal generators with known, controllable ground truth, intended both as hypothesis testbeds (which mechanisms produce which statistics?) and as calibration phantoms for estimators (which LRD/criticality estimator survives heavy tails and structural breaks?).

This review assembles the theoretical foundations of that programme. Each section states the mathematics the code implements, reviews the supporting literature, and closes with the implications for the library. Citations were collected and verified programmatically; residual verification caveats are listed in Appendix A.

Organisation. Section 2: fractional calculus, Mittag-Leffler relaxation, Matignon stability, and numerical schemes. Section 3: reservoir computing, the echo state property, memory capacity, and the small but growing fractional-RC literature. Section 4: neural-mass and neural-field models and their fractional-order generalisations. Section 5: neuronal avalanches, SOC, quasi-criticality, homeostasis, and E/I balance. Section 6: self-similar processes, LRD estimators, heavy tails, and the neural evidence. Section 7: how the pieces assemble in **fracres**, and the open research questions.

2 Fractional calculus and fractional-order systems

2.1 Definitions: Riemann–Liouville, Caputo, Grünwald–Letnikov

Fractional calculus extends differentiation and integration to non-integer order; the modern standard references are Podlubny [60], Kilbas, Srivastava & Trujillo [38], and the classic treatment of Oldham & Spanier [56]. For $0 < \alpha < 1$ the two derivatives used in **fracres** are the *Caputo* derivative

$${}^C D^\alpha f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{f'(\tau)}{(t-\tau)^\alpha} d\tau, \quad (1)$$

which respects classical initial conditions $f(0)$, and the *Grünwald–Letnikov* (GL) derivative, defined directly as a limit of weighted differences,

$$D^\alpha f(t) = \lim_{h \rightarrow 0} \frac{1}{h^\alpha} \sum_{j=0}^{\lfloor t/h \rfloor} c_j^{(\alpha)} f(t-jh), \quad c_0 = 1, \quad c_j = \left(1 - \frac{1+\alpha}{j}\right) c_{j-1}, \quad (2)$$

whose generalised-binomial weights $c_j^{(\alpha)}$ decay as a power law $j^{-\alpha-1}$ [56, 67]. The contrast with integer order is the point: a first derivative is *local* (two samples), while a fractional derivative is a *convolution over the whole history* with a power-law kernel — non-Markovian memory is definitional, not engineered. For power laws both derivatives agree with the closed form

$$D^\alpha t^\beta = \frac{\Gamma(\beta+1)}{\Gamma(\beta+1-\alpha)} t^{\beta-\alpha}, \quad \beta > -1, \quad (3)$$

[60, Eq. 2.117], which **fracres** uses as ground truth to validate its discrete kernels.

2.2 Mittag-Leffler relaxation

The eigenfunction of the Caputo derivative is the Mittag-Leffler function $E_\alpha(z) = \sum_{k \geq 0} z^k / \Gamma(\alpha k + 1)$: $x(t) = E_\alpha(\lambda t^\alpha)$ solves ${}^C D^\alpha x = \lambda x$, $x(0) = 1$ [26, 60]. Its asymptotics interpolate between a fast (stretched-exponential) initial decay and a power-law tail $x(t) \sim t^{-\alpha}$ [26, 30]: the impulse response of a fractional-order system *is* the non-exponential fading-memory profile that neural LRD demands, in contrast to the exponential forgetting of integer-order reservoirs (Section 3). The Caputo/Riemann–Liouville distinction matters in practice: with $x(0) \neq 0$, ${}^C D^\alpha f = D_{\text{RL}}^\alpha(f - f(0))$, a correction the library’s validation suite exercises explicitly on the Mittag-Leffler eigenfunction.

2.3 Stability: Matignon’s theorem

For the commensurate linear fractional system $D^\alpha x = Ax$, $0 < \alpha < 1$, the stability criterion replaces the classical left-half-plane condition by an *argument* condition:

Theorem 2.1 (Matignon [50]). *The system $D^\alpha x = Ax$ is asymptotically stable if and only if every eigenvalue λ_i of A satisfies*

$$|\arg(\lambda_i)| > \frac{\alpha\pi}{2}. \quad (4)$$

The unstable set is the cone of half-angle $\alpha\pi/2$ about the positive real axis. As $\alpha \rightarrow 1$ it fills the right half-plane, recovering the classical condition; for $\alpha < 1$ it *shrinks*, so a fractional-order system tolerates eigenvalues with positive real part — a strictly larger stable region [50, 51]. This is the fractional analogue of the echo-state spectral condition (Section 3): it defines the wedge that the reservoir Jacobian $A = W_{\text{res}} - \lambda I$ must respect, and it is the rigorous sense in which a fractional reservoir can sit closer to criticality — with a “hotter” connectome — than its integer-order counterpart. Matignon’s 1998 extension covers non-commensurate (multi-order) systems [51], the relevant theory once nodes are allowed heterogeneous orders α_i .

2.4 Numerical schemes

Two discretisations are implemented and validated in **fracres**. The *GL scheme* (2) is first-order, $O(h)$ [56, 67]; the *L1 scheme* for the Caputo derivative interpolates f' piecewise-linearly on each subinterval, giving weights $b_m = (m+1)^{1-\alpha} - m^{1-\alpha}$ and accuracy $O(h^{2-\alpha})$ [40]. The library’s measured convergence orders match both rates exactly (GL ≈ 1.00 ; L1 $\approx 2 - \alpha$ for $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$), confirming the implementations against (3). For nonlinear node dynamics, the fractional Adams–Bashforth–Moulton predictor–corrector of Diethelm, Ford & Freed is the standard higher-fidelity alternative [22]; the engineering literature also supplies rational (Oustaloup/CRONE) approximations used to realise fractional operators in analogue and neuromorphic hardware [53, 59]. All schemes face the same practical issue: the full-history convolution is $O(TL)$, and truncation (the short-memory principle [67]) trades accuracy for cost — for hardware, the retained history at tolerance ε scales as $\varepsilon^{-1/\alpha}$ [71].

2.5 Fractional-order reservoir computing

The fractional-RC literature is small but direct. Yao & Wang introduced the first fractional-order echo state network, updating reservoir states through a GL-type fractional difference and reporting improved prediction on chaotic benchmarks [80]. Goswami et al. prove the key theoretical separation: standard ESNs induce *short-memory* processes, while a fractional-difference reservoir generates polynomially decaying autocorrelation consistent with statistical long memory [27] — precisely the property required to model LRD in neural signals. A 2026 review positions fractional-order RC as a general extension of the RC paradigm, covering the memory-capacity/fading-memory trade-off controlled by the order α [49], and Teuscher’s hardware analysis quantifies what physical fractional devices can deliver [71]. Notably absent from the literature: any fractional reservoir targeted at *critical* (avalanche) statistics, any coupling of fractional nodes to neural-mass/field equations, and any stochastic-drive (fGn) formulation — the space **fracres** occupies.

Implications for fracres. The GL/L1 kernels (`kernels.py`) are validated against (3) and the Mittag-Leffler eigenfunction; Matignon’s criterion (4) is implemented as the edge-of-chaos control (`stability.py`); the demonstrated “fractional advantage” — one connectome stable for $\alpha < 0.84$ where the $\alpha = 1$ ESN diverges — is a direct numerical exhibition of Theorem 4.

3 Reservoir computing

3.1 The reservoir/readout split

Reservoir computing trains only a linear readout of a fixed, randomly wired dynamical system. The two founding formulations are Jaeger’s echo state networks (ESNs) [33, 34] and Maass, Natschläger & Markram’s liquid state machines (LSMs) [46], shown to be instances of one framework by Verstraeten et al. [74] and surveyed by Lukoševičius & Jaeger [44]. Two ideas from the foundations carry directly into the present programme. First, the LSM’s thesis that cortical microcircuits compute on *transients* rather than attractors [46] aligns with critical, avalanche-like dynamics. Second, the essential object is the reservoir/readout split itself, not the neuron model [74] — so replacing the node’s dynamical law (integer $\rightarrow$ fractional) while keeping the split is a legitimate, minimal modification, which is what **fracres** does.

3.2 Echo state property and fading memory

A reservoir has the *echo state property* (ESP) if its state is uniquely determined by the left-infinite input history [33]; the classical sufficient conditions bound the spectral radius (Jaeger) or, more tightly, the largest singular value of W_{res} [13]. Yildiz, Jaeger & Kiebel showed the ESP is input-dependent: for some inputs no scaling of a given reservoir has echo states [81]. Both theorems assume *geometric* contraction — their Lipschitz bounds do not directly apply to power-law memory, and re-establishing an ESP analogue under algebraic decay is an open theoretical task for fractional RC. In **fracres** the role of the contraction is played by the gain-normalised fractional update (unit linear memory gain) plus the leak $-\lambda x$, with Matignon’s wedge as the linearised stability certificate; a formal fractional ESP proof remains future work (`goswami2026longmemory` proves the complementary result that fractional reservoirs *do* produce long-memory stationary states [27]).

3.3 Memory capacity

Jaeger’s memory-capacity curve measures how well the readout reconstructs delayed inputs $u_{t-\tau}$; linear ESNs with N units have total capacity $\leq N$, with recall decaying *exponentially* in the delay [35]. Dambre et al. generalised this to a full information-processing capacity over all

polynomial functionals of the input, proving a conservation law: total capacity is bounded by the number of independent reservoir variables [18]. The design question for fractional RC is therefore not “more capacity” but *redistribution*: power-law kernels trade short-delay recall for algebraically decaying long-delay recall within the same budget. The delayed-copy benchmarks in **fracres** (test correlation $0.82 \rightarrow 0.51$ over delays $1 \rightarrow 10$, and $0.64 \rightarrow 0.95$ after the (h, λ) tuning study) measure exactly this curve.

3.4 Edge of chaos

Bertschinger & Natschläger showed that real-time reservoir computation peaks at the transition between ordered and chaotic dynamics — the *edge of chaos* [10]; Legenstein & Maass extended the analysis to biologically realistic circuits and proposed predicting performance from the Jacobian spectrum [39]. In classical RC the only route to the edge is tuning the spectral radius $\rho(W_{\text{res}}) \rightarrow 1$. A fractional reservoir has a second axis: the order α moves the Matignon wedge (4), so the *same* connectome can be unstable at $\alpha = 1$ and stable at $\alpha < 1$. Criticality tuning and memory tuning partially decouple — a structural advantage that the classical framework lacks.

3.5 Modern context

Physical and delay-based reservoirs replace topological memory with dynamical memory in the substrate itself [55, 69] — the closest classical analogue of the fractional design. Next-generation reservoir computing strips the reservoir to its memory-functional content (nonlinear vector autoregression) [24]: both a competitor and a natural host for fractional (long-memory) feature maps.

Implications for fracres. The library keeps the canonical split (frozen reservoir, trained linear readout, ridge or gradient) and the delayed-copy memory probe, but replaces the exponential-memory node with a fractional one and the spectral-radius edge-of-chaos knob with the Matignon wedge. The open theory items are a fractional ESP theorem and a capacity-redistribution theorem for power-law kernels.

4 Neural mass and neural field models

4.1 Classical models

The biological substrate of the phantom brain comes from two modelling traditions. *Neural masses* coarse-grain a cortical column into coupled excitatory/inhibitory populations: Wilson & Cowan’s founding model [78, 79] with its sigmoidal response and E/I loop producing multi-stability and oscillations; the thalamocortical alpha-rhythm loop of Lopes da Silva et al. [43]; and the Jansen–Rit column, the standard EEG/VEP generator [36]. *Neural fields* replace the lumped populations by a continuum sheet with distance-dependent connectivity: Amari’s analysis of lateral-inhibition (Mexican-hat) fields, proving existence and stability of localised bump solutions [3], systematised in modern reviews [16, 17, 23] and situated in the model hierarchy by Deco et al. [20]. Pattern formation via short-range excitation/long-range inhibition is the same activator–inhibitor (Turing) mechanism as in reaction–diffusion theory [54].

A unifying observation, due to Deco et al. [20]: every rung of this reduction ladder assumes *exponential* memory kernels (second-order synaptic filters, exponential adaptation). Fractional calculus replaces them with power-law kernels at the cost of a single order parameter — the minimal generalisation that can express the scale-free temporal filtering neural tissue actually exhibits.

4.2 Fractional evidence at the cellular level

The strongest empirical motivation is cellular. Lundstrom, Higgs, Spain & Fairhall showed that neocortical pyramidal neurons perform *fractional-order differentiation*: spike-frequency adaptation to slowly varying input follows a fractional derivative, with memory spanning seconds to hundreds of seconds [45]. Anastasio had earlier shown the vestibulo-oculomotor integrator behaves as a fractional-order integrator [4]. If single neurons already compute fractional operators, fractional-order *population* models are not an abstraction but a coarse-graining of measured physiology.

4.3 Fractional-order population models

The fractional generalisation programme is underway but young: Mondal et al. analyse a fractional Wilson–Cowan E/I network, showing the derivative order acts as an extra bifurcation parameter enriching the dynamical repertoire [52]; González-Ramírez formulates a Caputo neural field with lateral-inhibition kernel and derives approximate travelling waves whose speed and profile the order modulates [25]; and the fractional recurrent-network literature supplies Mittag-Leffler stability theory for fractional Hopfield and FitzHugh–Nagumo systems [12, 76]. A genuine gap: no fractional-order Jansen–Rit/EEG neural-mass model appears in the indexed literature — the closest items are the fractional field [25] and fractional RNN work [76].

Implications for *fracres*. The library implements the fractional Wilson–Cowan reservoir (stacked E/I masses, separate $\tau_E^\alpha, \tau_I^\alpha$) and the fractional Amari field (Mexican-hat ring connectivity, nonlinearity inside the convolution $w * S(u)$). The missing fractional Jansen–Rit/EEG mass model is a positioning opportunity for the EEG-fitting phase of the programme.

5 Critical brain dynamics

5.1 Neuronal avalanches

Beggs & Plenz reported that spontaneous LFP events in cortical slice cultures distribute in size as a power law $P(S) \sim S^{-\tau}$ with $\tau \approx 1.5$, and in duration with $\alpha \approx 2.0$ — the mean-field exponents of a critical branching process (branching ratio $\sigma = 1$) [6]; the patterns are diverse, precise, and reproducible over hours [7]. The exponents obey the crackling-noise relation $\gamma = (\alpha - 1)/(\tau - 1) \approx 2$, a non-trivial consistency check on the critical branching picture [6]. Reviews have elevated avalanches to a candidate organising principle of cortical activity [58] and catalogued the functional benefits of the critical point — maximal dynamic range, information capacity, and an optimal storage/transfer trade-off [14, 68]. Spike-avalanche analyses in vivo across species and states find near-universal exponents with state-dependent deviations [21, 64].

5.2 Scepticism and methodology

The criticality hypothesis has serious critics. Beggs & Timme catalogue the failure modes: alternative mechanisms mimicking power laws, thresholding sensitivity, and the rarity of proper statistical testing [8]. The methodological standard is Clauset, Shalizi & Newman: maximum-likelihood exponents with $x_{\min}$ selection, bootstrapped goodness-of-fit, and likelihood-ratio comparison against log-normal and exponential alternatives [15] — the estimator *fracres* implements for its avalanche exponents. A second pitfall is *subsampling*: any electrode array records a small fraction of the network, biasing measured distributions [63]; with subsampling-invariant analysis, in-vivo spike avalanches look *slightly subcritical with reverberation* rather than exactly critical [62].

5.3 SOC, quasi-SOC, and homeostasis

Self-organised criticality explains scale invariance without parameter tuning [5, 37], but its standard models require local conservation — which neural networks, creating and destroying activity, violate. Bonachela & Muñoz proved that feedback-driven non-conservative systems self-organise only to *apparent*, finite-range scale invariance: quasi-criticality [11]. Williams-García et al. located driven, resource-limited branching models on a nonequilibrium Widom line of near-critical susceptibility maxima, producing approximate power laws with near-canonical exponents without exact tuning [77]. The biological mechanism that could hold a network in this regime is homeostatic plasticity — synaptic scaling and intrinsic-excitability regulation stabilising average firing rates [72]. The modern consensus picture is therefore *quasi-criticality maintained by homeostatic control*, not strict SOC.

5.4 E/I balance

Excitation/inhibition balance is the empirically identified control parameter: pharmacological manipulation of inhibition moves avalanche duration exponents and waiting-time clustering while τ stays near 1.5 [42]; avalanches and long-range temporal correlations co-emerge in the same human MEG networks and covary with E/I-related state [21, 61]. This motivates both the E/I structure of the Wilson–Cowan reservoir and the qSOC homeostat’s placement as a threshold on population activity.

Implications for fracres. The qSOC reservoir implements exactly the quasi-criticality-with-homeostasis picture: a windowed-energy feedback pulling the network toward a critical set point [11, 72]. The library’s first criticality study found power-law-like avalanches ($\tau \approx 1.2$, duration exponents rising with the fractional order) whose statistics *survive* a strong perturbation only under homeostatic control — consistent with the homeostat’s role being robustness of the operating point rather than the source of the exponents. The methodological lessons of [15, 62] (MLE exponents, subsampling awareness) are built into `metrics.py`.

6 Long-range dependence: theory and estimation

6.1 Self-similar processes and the Hurst phenomenon

Empirical LRD began with Hurst’s rescaled-range analysis of Nile minima: $R/S \sim n^K$ with $K \approx 0.73 \neq 1/2$ [32]. Mandelbrot & Wallis separated the phenomenon into the *Joseph effect* (temporal persistence) and the *Noah effect* (heavy-tailed amplitudes) [48], and Mandelbrot & Van Ness supplied the canonical model: fractional Brownian motion $B_H(t)$, the self-similar Gaussian process with stationary increments, and its increment process fractional Gaussian noise (fGn), whose autocorrelation is non-summable for $H > 1/2$ [47]. The fGn autocovariance

$$r(k) = \frac{1}{2}(|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H}), \quad r(0) = 1, \quad (5)$$

is exactly what the Davies–Harte circulant-embedding algorithm synthesises in $O(n \log n)$ [19] — the drive generator in `fracres` (`drivers.py`), scaled so the realised process has unit variance in expectation and the exact $r(k)$. ARFIMA/fractional differencing $(1 - B)^d$ provides the discrete-time sibling with $\beta = 2d$ [28].

6.2 Estimators and their failure modes

Second-order LRD theory defines long memory equivalently by non-summable covariance, a spectral pole $f(\lambda) \sim |\lambda|^{-2d}$, or aggregated variance $\text{Var}(X^{(m)}) \sim m^{2H-2}$ [9]. The estimator landscape is treacherous: Taqqu, Teverovsky & Willinger’s Monte Carlo study showed most

H estimators are strongly biased at realistic lengths, so agreement among methods is required before claiming LRD [70]. The two estimators in **fracres** cover the time and frequency domains: detrended fluctuation analysis [57], for which the DFA exponent equals H for stationary fGn (and $H + 1$ for fBm, whence a stationarity check), and spectral-slope estimation, since the fGn PSD decays as

$$S(f) \propto f^{-\beta}, \quad \beta = 2H - 1 \quad (6)$$

over the scaling range — so $1/f$ -like spectra ($\beta \approx 1$) sit at the edge of the fGn family. Wavelet log-scale diagrams estimate H from the scale-energy law $\mathbb{E}|d_{j,k}|^2 \propto 2^{j(2H)}$, exploiting the decorrelation of wavelet coefficients for fGn-like processes [1, 73]; the wavelet lens connects self-similarity to Besov regularity [2], the function-space framing the library’s regulariser adopts.

6.3 Heavy tails: the variance trap

All of the above presupposes finite variance. For α -stable processes with tail index $\alpha_{\text{tail}} < 2$, variance is infinite, covariance undefined, and *every* L^2 -based estimator — R/S, variance–time, aggregated variance, DFA to second order — is biased or meaningless: heavy tails masquerade as, and corrupt estimates of, long memory [65, 66]. Samorodnitsky’s survey shows LRD is *plural* under heavy tails: scaling of sums, maxima, and ergodic structure need not agree, and dependence must be characterised by moment-free measures [65]. The principled response is to move from covariance to *scale-space regularity*: p -th order wavelet/Besov energies with $p < \alpha_{\text{tail}}$, where the smoothness index must respect $s < \min\{H, 1/p\}$ — the $1/p$ ceiling is the jump/heavy-tail guardrail. This is exactly the Besov index convention implemented in **fracres**’ `training.besov_indices` and its Littlewood–Paley regulariser, and it frames the library’s thesis-level claim: covariance-LRD is the $p = 2$, stationary shadow of a broader Besov–wavelet scaling phenomenon.

6.4 LRD in neural signals

The neural evidence is strong and old: amplitude envelopes of spontaneous human MEG/EEG oscillations show power-law correlations over hundreds of seconds with DFA exponents ≈ 0.6 – 0.8 [41]; $1/f$ aperiodic spectra are ubiquitous across recording modalities and covary with E/I balance, state, and age [31, 75]; and DFA methodology — including the $\alpha = H$ vs. $H + 1$ ambiguity, crossovers, and nonstationarity pitfalls — is codified for neurophysiology by Hardstone et al. [29], who also survey DFA changes in sleep, development, and pathology. These are precisely the statistics a phantom brain must reproduce, and the estimators it must calibrate against.

Implications for **fracres.** The library’s metric suite (`metrics.py`) implements DFA and the log-binned low-frequency spectral fit with the finite-sample safeguards the literature demands [29, 70]; the drive is exact Davies–Harte fGn [19]; and the Besov regulariser enforces the heavy-tail-aware bounds $p < \alpha_{\text{tail}}$, $s < \min\{H, 1/p\}$ [2, 65]. The remaining gap is a non-Gaussian (stable) drive so the $\alpha_{\text{tail}} < 2$ machinery is actually exercised.

7 Synthesis: the **fracres** programme in context

Table 1 summarises how each body of theory lands in code. Three conclusions emerge from the assembled literature.

1. The niche is genuinely open. Fractional RC exists but is four papers deep and none of it targets critical statistics, neural-mass structure, or stochastic fGn drive [27, 49, 71, 80]. Fractional neural models exist but no fractional Jansen–Rit/EEG mass model is indexed (Section 4). The intersection — fractional-node reservoirs as *critical brain phantoms* — appears unoccupied.

Table 1: Theory-to-code map for **fracres**, with the anchoring literature.

Component	Theory	Key references
<code>kernels.py</code>	GL / L1-Caputo schemes	[40, 56, 67]
<code>validation.py</code>	$D^\alpha t^\beta$, Mittag-Leffler	[26, 60]
<code>stability.py</code>	Matignon wedge, edge of chaos	[10, 50]
<code>reservoirs.py</code>	ESN split, fractional Wilson–Cowan/Amari	[3, 33, 79]
<code>qSOC homeostat</code>	Quasi-criticality + homeostasis	[11, 72]
<code>drivers.py</code>	fGn, Davies–Harte synthesis	[19, 47]
<code>regularizers.py</code>	Besov / Littlewood–Paley	[2, 65]
<code>training.py</code>	Readout-only ridge/gradient	[33, 74]
<code>metrics.py</code>	DFA, spectral slope, CSN MLE	[15, 57]

2. The biological and theoretical motivations are independent and convergent. Cellular fractional computation [4, 45], power-law population filtering (Section 4), quasi-criticality with homeostasis [11, 72], and the failure of L^2 LRD theory under heavy tails [65] each independently push toward the same construction: power-law memory operators, criticality control, and Besov-geometry regularisation.

3. The sharpest open problems are theoretical, and they are named. (a) A fractional echo-state property: contractivity of the fractional state map under algebraic (non-geometric) decay, in the spirit of [13, 81]. (b) Memory-capacity redistribution: how power-law kernels reallocate the Dambre capacity budget across delays [18]. (c) Criticality claims must be made with Clauset–Shalizi–Newman rigour and subsampling awareness [15, 62] — avalanches from a phantom are hypotheses about mechanism, not evidence about brains, until confronted with data. (d) The inverse problem: inferring (α, H) and the E/I operating point from real EEG, where the estimator pathologies of Section 6 are themselves part of the question.

A Bibliographic verification

All 81 references were collected and verified programmatically (September 2026) against live bibliographic APIs — Crossref/doi.org content negotiation, OpenAlex, OpenLibrary, NeurIPS proceedings, and arXiv abstract pages — with per-entry verification logs in `docs/litreview/notes/`. Residual caveats: Matignon 1996 [50] has no DOI (OpenAlex-verified; page numbers from secondary citations); the Jaeger GMD reports [33, 35] are verified via author-hosted copies; the Abry–Flandrin–Taqu–Veitch chapter [2] is verified at container level only; Mandelbrot & Wallis is cited per verified publisher metadata (WRR 4(5), 1968), which corrects the widespread miscitation as vol. 5, 1969 [48]; Voytek & Knight 2015 appears in *Biological Psychiatry*, not *Neuron* [75].

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